

Human review packet: GCG24UserItemFairness

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: GCG24UserItemFairness

Paper id: GCG24UserItemFairness

Source version: NeurIPS 2024 paper; byte-pinned publication source record

Source record: audit/paper_statement_map.json

Rows: 37

Generated: 2026-08-19

Source URL: [official source](#)

Review status: 0/37 source-claim screens; 0/15 library prerequisite screens. This is a review worksheet while those direct checks remain pending; it is not a final validation receipt.

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) EconCSLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-EconCSLib-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper GCG24UserItemFairness --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's reviewer-owned review record.

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Material library prerequisites

EconCSLib.Policy

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\alpha : \text{Type } u_1) \rightarrow (\beta : \text{Type } u_2) \rightarrow \alpha \rightarrow \text{PMF } \beta$

Exact Lean library declaration

abbrev Policy ($\alpha \beta : \text{Type}^*$) := $\alpha \rightarrow \text{PMF } \beta$

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Policy.FiberRepresentatives

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$\{\alpha : \text{Type } u_1\} \rightarrow \{\kappa : \text{Type } u_2\} \rightarrow (\alpha \rightarrow \kappa) \rightarrow \text{Type } (\max u_1 u_2)$

Exact Lean library declaration

```
structure FiberRepresentatives {α κ : Type*} (τ : α → κ) where
  repr : κ → α
  repr_spec : ∀ k, τ (repr k) = k
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Policy.FiberRepresentatives.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{α : Type u_1} →  
  {κ : Type u_2} → {τ : α → κ} → (repr : κ → α) → (∀ (k : κ), τ (repr k) = k) → EconCSLib.Policy.FiberRepresentatives τ
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Policy.FiberRepresentatives.ofSurjective

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
{κ : Type u_2} → {τ : α → κ} → (hτ : Function.Surjective τ) → { repr := fun k => Classical.choose ⋯, repr_spec := ⋯ }
```

Exact Lean library declaration

```
noncomputable def FiberRepresentatives.ofSurjective {α κ : Type*}  
  {τ : α → κ} (hτ : Function.Surjective τ) : FiberRepresentatives τ where  
  repr := fun k => Classical.choose (hτ k)  
  repr_spec := fun k => Classical.choose_spec (hτ k)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Policy.FiberRepresentatives.repr

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{\alpha : \text{Type } u_1\} \rightarrow \{\kappa : \text{Type } u_2\} \rightarrow \{\tau : \alpha \rightarrow \kappa\} \rightarrow (\text{self} : \text{EconCSLib.Policy.FiberRepresentatives } \tau) \rightarrow (a : \kappa) \rightarrow \text{self}.1 a$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Policy.activePairs

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
{β : Type u_2} →  
  [inst : Fintype α] →  
    [DecidableEq α] → [inst_2 : Fintype β] → [DecidableEq β] → (ρ : EconCSLib.Policy α β) → {p | (ρ p.1) p.2 ≠ 0}
```

Exact Lean library declaration

```
noncomputable def activePairs {α β : Type*}  
  [Fintype α] [DecidableEq α] [Fintype β] [DecidableEq β]  
  (ρ : Policy α β) : Finset (α × β) :=  
  Finset.univ.filter fun p => ρ p.1 p.2 ≠ 0
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Policy.activePairsCard

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
{β : Type u_2} →  
  [inst : Fintype α] →  
    [inst_1 : DecidableEq α] →  
      [inst_2 : Fintype β] → [inst_3 : DecidableEq β] → (ρ : EconCSLib.Policy α β) → ρ.activePairs.card
```

Exact Lean library declaration

```
noncomputable def activePairsCard {α β : Type*}  
  [Fintype α] [DecidableEq α] [Fintype β] [DecidableEq β]  
  (ρ : Policy α β) : ℕ :=  
  (activePairs ρ).card
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfExp

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{\alpha : \text{Type } u_1\} \rightarrow [\text{inst} : \text{Fintype } \alpha] \rightarrow [\text{DecidableEq } \alpha] \rightarrow (\mu : \text{PMF } \alpha) \rightarrow (f : \alpha \rightarrow \mathbb{R}) \rightarrow \sum a, (\mu a).toReal * f a$

Exact Lean library declaration

```
noncomputable def pmfExp {α : Type*} [Fintype α] [DecidableEq α]
  (μ : PMF α) (f : α → ℝ) : ℝ :=
  ∑ a, (μ a).toReal * f a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Policy.agentScore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
{β : Type u_2} →  
  [inst : Fintype β] →  
    [inst_1 : DecidableEq β] →  
      (p : EconCSLib.Policy α β) → (score : α → β → ℝ) → (a : α) → EconCSLib.pmfExp (p a) (score a)
```

Exact Lean library declaration

```
noncomputable def agentScore {α β : Type*}  
  [Fintype β] [DecidableEq β]  
  (p : Policy α β) (score : α → β → ℝ) (a : α) : ℝ :=  
  pmfExp (p a) (score a)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Policy.average0n

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
{β : Type u_2} →  
  [inst : Fintype β] →  
  (p : α → PMF β) → (s : Finset α) → (hs : s.Nonempty) → PMF.ofFintype (fun b => (∑ a ∈ s, (p a) b) / ↑s.card) ...
```

Exact Lean library declaration

```
noncomputable def average0n {α β : Type*} [Fintype β]  
  (p : α → PMF β) (s : Finset α) (hs : s.Nonempty) : PMF β :=  
  PMF.ofFintype  
    (fun b : β => (∑ a ∈ s, p a b) / (s.card : ENNReal))  
  (by  
    classical  
    have hcard_ne_zero_nat : s.card ≠ 0 := Finset.card_ne_zero.mpr hs  
    have hcard_ne_zero : (s.card : ENNReal) ≠ 0 := by  
      exact_mod_cast hcard_ne_zero_nat  
    have hcard_ne_top : (s.card : ENNReal) ≠ ⊤ :=  
      ENNReal.natCast_ne_top s.card  
    calc  
      ∑ b : β, (∑ a ∈ s, p a b) / (s.card : ENNReal)  
      = ∑ b : β, (s.card : ENNReal)-1 * (∑ a ∈ s, p a b) := by  
        refine Finset.sum_congr rfl ?_  
        intro b  
        rw [ENNReal.div_eq_inv_mul]  
      _ = (s.card : ENNReal)-1 * (∑ b : β, ∑ a ∈ s, p a b) := by  
        rw [Finset.mul_sum]  
      _ = (s.card : ENNReal)-1 * (∑ a ∈ s, ∑ b : β, p a b) := by  
        rw [Finset.sum_comm]  
      _ = (s.card : ENNReal)-1 * (∑ a ∈ s, (1 : ENNReal)) := by  
        refine congrArg (fun z => (s.card : ENNReal)-1 * z) ?_  
        refine Finset.sum_congr rfl ?_  
        intro a  
        simp [tsum_fintype] using (PMF.tsum_coe (p a))  
      _ = (s.card : ENNReal)-1 * (s.card : ENNReal) := by  
        simp  
      _ = 1 := ENNReal.inv_mul_cancel hcard_ne_zero hcard_ne_top)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Policy.descendAlong

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
  {κ : Type u_2} →  
    {β : Type u_3} →  
      {τ : α → κ} →  
        (reprs : EconCSLib.Policy.FiberRepresentatives τ) → (p : EconCSLib.Policy α β) → (a : κ) → p (reprs.repr a)
```

Exact Lean library declaration

```
def descendAlong {α κ β : Type*} {τ : α → κ}  
  (reprs : FiberRepresentatives τ) (p : Policy α β) : Policy κ β :=  
  fun k => p (reprs.repr k)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Policy.liftAlong

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{\alpha : \text{Type } u_1\} \rightarrow \{\kappa : \text{Type } u_2\} \rightarrow \{\beta : \text{Type } u_3\} \rightarrow (\tau : \alpha \rightarrow \kappa) \rightarrow (\rho : \text{EconCSLib.Policy } \kappa \beta) \rightarrow (a : \alpha) \rightarrow \rho (\tau a)$

Exact Lean library declaration

```
def liftAlong {α κ β : Type*} (τ : α → κ) (ρ : Policy κ β) : Policy α β :=  
  fun a => ρ (τ a)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Policy.multiAssignedActions

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →
{β : Type u_2} →
  [inst : Fintype α] →
    [inst_1 : DecidableEq α] →
      [inst_2 : Fintype β] →
        [DecidableEq β] → (p : EconCSLib.Policy α β) → {b | ∃ a a', a ≠ a' ∧ (p a) b ≠ 0 ∧ (p a') b ≠ 0}
```

Exact Lean library declaration

```
noncomputable def multiAssignedActions {α β : Type*}
  [Fintype α] [DecidableEq α] [Fintype β] [DecidableEq β]
  (p : Policy α β) : Finset β :=
  Finset.univ.filter fun b => ∃ a a', a ≠ a' ∧ p a b ≠ 0 ∧ p a' b ≠ 0
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.finiteMax

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} → [inst : Fintype α] → [inst_1 : Nonempty α] → (f : α → ℝ) → Finset.univ.sup' ... f
```

Exact Lean library declaration

```
def finiteMax {α : Type*} [Fintype α] [Nonempty α]  
  (f : α → ℝ) : ℝ :=  
  (Finset.univ : Finset α).sup' Finset.univ_nonempty f
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.finiteMin

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} → [inst : Fintype α] → [inst_1 : Nonempty α] → (f : α → ℝ) → Finset.univ.inf' ... f
```

Exact Lean library declaration

```
def finiteMin {α : Type*} [Fintype α] [Nonempty α]  
  (f : α → ℝ) : ℝ :=  
  (Finset.univ : Finset α).inf' Finset.univ_nonempty f
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 149-150

Verbatim source input

Given a recommendation policy ρ , user fairness is quantified as the minimum normalized user utility $U_{\min}(\rho, w) = \min_i U_i(\rho, w)$, and item fairness analogously as $I_{\min}(\rho, w) = \min_j I_j(\rho, w)$.

Expanded PaperInterface specification

```
∀ {m n : ℕ} [inst : NeZero n] (W : GCG24UserItemFairness.RecommendationModel m n)
  (ρ : GCG24UserItemFairness.Policy m n),
  GCG24UserItemFairness.ProofBridge.itemFairness W ρ =
    EconCSLib.finiteMin (GCG24UserItemFairness.ProofBridge.normalizedItemUtility W ρ)
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.itemFairness_realizes_spec

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 134-145

Verbatim source input

recommendation policy ρ , let $U_i(\rho, w)$ be user i 's utility from ρ , normalized by the utility $\max_j w_{ij}$ they would receive from being recommended their best match. Let $I_j(\rho, w)$ be item j 's utility from ρ , normalized by the utility j receives if it is recommended to every user, i w_{ij} . The normalized utilities are:

$$U_i(\rho, w) = \frac{\sum_j \rho_{ij} w_{ij}}{\max_j w_{ij}}, I_j(\rho, w) = \frac{\sum_i \rho_{ij} w_{ij}}{\sum_i w_{ij}}$$

Expanded PaperInterface specification

$\forall \{m, n : \mathbb{N}\} (W : \text{GCG24UserItemFairness.RecommendationModel } m \ n) (j : \text{GCG24UserItemFairness.Item } n),$
 $\text{GCG24UserItemFairness.ProofBridge.itemNormalizer } W \ j = \sum u, W.\text{utility } u \ j$

Lean proof endpoint (built separately)

`GCG24UserItemFairness.PaperInterface.itemNormalizer_realizes_spec`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 134-145

Verbatim source input

recommendation policy ρ , let $U_i(\rho, w)$ be user i 's utility from ρ , normalized by the utility $\max_j w_{ij}$ they would receive from being recommended their best match. Let $I_j(\rho, w)$ be item j 's utility from ρ , normalized by the utility j receives if it is recommended to every user, i w_{ij} . The normalized utilities are:

$$U_i(\rho, w) = \frac{\sum_j \rho_{ij} w_{ij}}{\max_j w_{ij}}, I_j(\rho, w) = \frac{\sum_i \rho_{ij} w_{ij}}{\sum_i w_{ij}}$$

Expanded PaperInterface specification

$\forall \{m, n : \mathbb{N}\} (W : \text{GCG24UserItemFairness.RecommendationModel } m \ n) (\rho : \text{GCG24UserItemFairness.Policy } m \ n) (j : \text{GCG24UserItemFairness.Item } n),$
 $\text{GCG24UserItemFairness.ProofBridge.normalizedItemUtility } W \ \rho \ j =$
 $\text{GCG24UserItemFairness.ProofBridge.rawItemUtility } W \ \rho \ j / \text{GCG24UserItemFairness.ProofBridge.itemNormalizer } W \ j$

Lean proof endpoint (built separately)

`GCG24UserItemFairness.PaperInterface.normalizedItemUtility_realizes_spec`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 134-145

Verbatim source input

recommendation policy ρ , let $U_i(\rho, w)$ be user i 's utility from ρ , normalized by the utility $\max_j w_{ij}$ they would receive from being recommended their best match. Let $I_j(\rho, w)$ be item j 's utility from ρ , normalized by the utility j receives if it is recommended to every user, i w_{ij} . The normalized utilities are:

$$U_i(\rho, w) = \frac{\sum_j \rho_{ij} w_{ij}}{\max_j w_{ij}}, \quad I_j(\rho, w) = \frac{\sum_i \rho_{ij} w_{ij}}{\sum_i w_{ij}}$$

Expanded PaperInterface specification

```
∀ {m n : ℕ} [inst : NeZero n] (W : GCG24UserItemFairness.RecommendationModel m n) (ρ : GCG24UserItemFairness.Policy m n)
  (u : GCG24UserItemFairness.User m),
  GCG24UserItemFairness.ProofBridge.normalizedUserUtility W ρ u =
    GCG24UserItemFairness.ProofBridge.rawUserUtility W ρ u / W.bestItemUtility u
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.normalizedUserUtility_realizes_spec

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 174-199

Verbatim source input

Research questions: price of fairness and misestimation. We can now define the price of item fairness on user fairness π_U
|I (“price of fairness”) as the decrease in user fairness with maximal item fairness constraints:
*
*
Umin
(w) - Umin
($y = 1, w$)
F
 π_U
.
|I (w) :=
* (w)
Umin
F
We ask how π_U
|I changes with the utility matrix w. We note that while this question (in terms of solutions to Problem (1)) can be simply stated, as we detail in Section 3, finding a closed form expression for Umin
(1, w) – the optimal minimum normalized user utility given item fairness constraints – in terms of w is theoretically challenging.

Similarly, we investigate the price of misestimation. Let $\hat{w}_{ij}$ denote the platform’s estimate of the

Expanded PaperInterface specification

```
∀ {m n : ℕ} [inst : NeZero m] [inst_1 : NeZero n] (W : GCG24UserItemFairness.RecommendationModel m n),  
  GCG24UserItemFairness.ProofBridge.priceOfFairness W = GCG24UserItemFairness.ProofBridge.priceOfFairnessAt W 1
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.priceOfFairness_realizes_spec

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 199-220

Verbatim source input

Similarly, we investigate the price of misestimation. Let $\hat{w}_{ij}$ denote the platform's estimate of the utility of recommending i to j ; let $p(y)$ be a policy that solves the optimization problem above with

[form-feed in source extraction]

*
misestimated utilities, that is, $p(y)$ attains $U_{\min}(y, \hat{w})$.¹ Then we define the price of misestimation
M
on user utility ("price of misestimation") π_U as
M
 $(y, w, \hat{w}) :=$
 π_U

*
 $U_{\min}(y, w) - U_{\min}(p(y), w)$
,
* (y, w)
 $U_{\min}$

that is, the decrease in true user fairness due to optimizing with estimated utilities. In particular, in Section 5 we will examine whether item fairness exacerbates the price of misestimation by comparing

Expanded PaperInterface specification

```
∀ {m n : ℕ} [inst : NeZero m] [inst_1 : NeZero n] (E : GCG24UserItemFairness.EstimatedRecommendationModel m n) (γ : ℝ)
  (phat : GCG24UserItemFairness.Policy m n)
  (hEstimatedOptimal : GCG24UserItemFairness.ProofBridge.solvesProblemOne E.estimatedModel γ phat),
  GCG24UserItemFairness.ProofBridge.priceOfMisestimation E γ phat hEstimatedOptimal =
    (E.trueModel.optimalUserFairnessAtLevel γ - E.trueModel.userFairness phat) /
      E.trueModel.optimalUserFairnessAtLevel γ
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.priceOfMisestimation_realizes_spec

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 3401-3412

Verbatim source input

First, we use an analogue of the sparsity result Proposition 2 to show that the optimal x, z has the following sparse structure.
Lemma 13. Let x, z be an optimal basic feasible solution to the linear program Problem 11. There
,
,
,
,
,
is some $j \leq n+1$
2 such that for all $j > j$, $x_j = 0$ and for all $j < j$, $z_j = z_{n-j} + 1 = 0$. Let the
pivot index t denote the minimum such j .
This sparse structure implies uniqueness.

Expanded PaperInterface specification

$\forall \{n : \mathbb{N}\} (\text{beta} : \mathbb{R}) (v : \text{GCG24UserItemFairness.Item } n \rightarrow \mathbb{R}) (p : \text{GCG24UserItemFairness.TypePolicy } 3 \ n) (\text{ell} : \mathbb{R}),$
GCG24UserItemFairness.ProofBridge.problem11BasicFeasible beta v p ell $\leftrightarrow$
GCG24UserItemFairness.ProofBridge.problem11PointOfPolicy p ell $\in$
Set.extremePoints $\mathbb{R}$ (GCG24UserItemFairness.ProofBridge.problem11EqualityFeasibleSet beta v)

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.problem11BasicFeasible_realizes_spec

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 3375-3400

Verbatim source input

$S' = \{x, y, z \in S_{\text{symm}} : x_j = y_{n-j+1}, z_j = z_{n-j+1} \text{ for all } j\}$.
Since y is completely determined by x , we can identify S' with Δ_{2n-1} and can identify $x, y, z \in S'$ simply by x, z .

*
Lemma 12. If a policy $\rho = (x, y, z) \in S_{\text{symm}}$ solves $U_{\min}(\rho) = U_{\min}$
(1), then there is some policy

*
 $\rho' = x', z' \in S'$ that solves $U_{\min}(\rho') = U_{\min}$
(1).

Applying the framework in Proposition 1 to this proof with $S = S'$ and using $\hat{w}$ as the user and item values, we reduce the problem to finding
 $\phi = \arg$

$\max$

$x, z \in \Delta_{2n-1}, \lambda$

λ
(11)

subject to $I_j(x, z) = \lambda$

$\forall j,$

and showing that this ϕ is unique.

Expanded PaperInterface specification

```
∀ {n : ℕ} (beta : ℝ) (v : GCG24UserItemFairness.Item n → ℝ),
  GCG24UserItemFairness.ProofBridge.problem11EqualityFeasibleSet beta v =
    {p |
      (∀ (j : GCG24UserItemFairness.Item n), 0 ≤ p.1.1 j) ∧
      (∀ (j : GCG24UserItemFairness.Item n), 0 ≤ p.1.2 j) ∧
      ∑ j, p.1.1 j = 1 ∧
      ∑ j, p.1.2 j = 1 ∧
      (∀ (j : GCG24UserItemFairness.Item n),
        p.1.2 (GCG24UserItemFairness.OpposingTypes.reverseItem j) = p.1.2 j) ∧
      ∀ (j : GCG24UserItemFairness.Item n),
        GCG24UserItemFairness.OpposingTypes.theorem4Problem11ItemValue beta v p.1.1 p.1.2 j = p.2}
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.problem11EqualityFeasibleSet_realizes_spec

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 292-316

Verbatim source input

Proposition 1. Suppose that for a set of recommendation policies $S \subseteq \Delta_m$

(i) S can be described by a finite set of linear constraints
(ii) There exists an optimal solution p^* to Problem (2) such that $p^* \in S$
(iii) p^* is the unique feasible solution to Problem (2) in S

*
Multiple policies p^* may optimize $U_{\min}$
($v, \hat{w}$), and different policies p^* may have different values of $U_{\min}(p; w)$. In our experimental results, we use the policy p^* found by the convex optimization solver.

1
4

[form-feed in source extraction]
Then, finding an optimal solution p^* to Problem (2) can be reduced to solving a linear program L .
With Proposition 1, the key technical challenges become: (a) given utility matrix w , finding S satisfying the above conditions; (b) given S and w , finding a closed-form expression for p^* ; (c) given
*
a closed form for p^* in terms of w , reasoning about the properties of item fair solution $U_{\min}(1, w)$.
To do this, for our main results, we construct S as a set of recommendations with a particular symmetric structure. Suppose users come in K types, where a user of type k shares a utility vector. Then, consider $S_{\text{symm}} = \{p : p_i = p_{i'} \text{ if } w_i = w_{i'}\} \subseteq \Delta_m$
n-1, the set of policies where users of the same type are given the same recommendation probabilities. Note that, in the extreme case, each user is their own type. Furthermore, let p_k denote the recommendation policy for type k , for $p \in S_{\text{symm}}$.

Expanded PaperInterface specification

```
∀ {m n : ℕ} [inst : NeZero m] [inst_1 : NeZero n] (W : GCG24UserItemFairness.RecommendationModel m n),
  GCG24UserItemFairness.assumption_positive_recommendation_utilities W →
  ∀ (S : Set (GCG24UserItemFairness.Policy m n))
    (D : GCG24UserItemFairness.RecommendationModel.FiniteLinearPolicySetDescription S)
    (pstar : GCG24UserItemFairness.Policy m n),
    W.IsOptimalAtLevel 1 pstar →
    pstar ∈ S →
    (∀ p ∈ S, W.itemFairness p = W.optimalItemFairness → p = pstar) →
    (W.PropositionOneLPFeasible D pstar W.optimalItemFairness ∧
     ∀ (p' : GCG24UserItemFairness.Policy m n) (ell' : ℝ),
       W.PropositionOneLPFeasible D p' ell' → ell' ≤ W.optimalItemFairness) ∧
    ∀ (p : GCG24UserItemFairness.Policy m n) (ell : ℝ),
      (W.PropositionOneLPFeasible D p ell ∧
       ∀ (p' : GCG24UserItemFairness.Policy m n) (ell' : ℝ),
         W.PropositionOneLPFeasible D p' ell' → ell' ≤ ell) →
      p = pstar ∧ ell = W.optimalItemFairness
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.proposition1_symmetric_lp_reduction

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 312-338

Verbatim source input

symmetric structure. Suppose users come in K types, where a user of type k shares a utility vector. Then, consider $S_{\text{symm}} = \{p : p_i = p_{i'} \text{ if } w_i = w_{i'}\} \subseteq \Delta_m$ $n-1$, the set of policies where users of the same type are given the same recommendation probabilities. Note that, in the extreme case, each user is their own type. Furthermore, let p_k denote the recommendation policy for type k , for $p \in S_{\text{symm}}$. Proposition 2. S_{symm} satisfies conditions (i) and (ii) in Proposition 1. Furthermore, solutions $p \in S_{\text{symm}}$ to Problem (2) have a sparse structure:

- If p is a basic feasible solution to the linear program L in Proposition 1, then there are at most $n + K - 1$ type-item pairs (k, j) such that $p_{kj} > 0$ (out of nK possible pairs).
- If p is also optimal, then there are at most $K - 1$ items that are ever recommended to more than one type of user, i.e., where $p_{kj}, p_{k'j} > 0$ for $k \neq k'$.

For our main results, we will leverage the sparsity structure in Proposition 2 to show, with further restrictions on utility matrix w , that S_{symm} also satisfies (iii). We will then derive closed-form expressions for that solution, and show how it changes as a function of w . The sparsity structure in Proposition 2 is also interesting in its own right. Without item fairness constraints, solutions will be highly sparse: each user will be recommended their most preferred item deterministically; each other item will have a recommendation probability of zero. Once we add item fairness constraints, solutions do not necessarily remain sparse. However, Proposition 2 shows that sparse solutions still arise under item fairness constraints in settings with symmetric solutions.

4

User preference diversity and fairness tradeoffs

We now use the theoretical framework developed in the above section to understand the effect of the structure of user utilities on the price of fairness. In particular, we identify “free fairness,” i.e., the price of fairness is low, when preferences are sufficiently diverse.

[Next verbatim source excerpt]

Proof. We will first prove that S_{symm} satisfies conditions (i) and (ii), and then show sparsity.

Part 1. Recall that for fixed w ,

\

$S_{\text{symm}} = \{p : p_i = p_{i'} \text{ if } w_i = w_{i'}\} =$

$\{p : p_i = p_{i'}\}$

$i, i' : w_i = w_{i'}$

which is a finite set of linear constraints, so condition (i) holds.

Now, we show condition (ii), that always there is some $\phi^* \in S_{\text{symm}}$ that is an optimal solution to Problem 1. Let p be an arbitrary optimal solution to Problem 1. Then p is also feasible: $I_{\min}(p) =$

*

$I_{\min}$

We first define a piece of convenient notation: if two users i, i' share the same utility vector $w_i = w_{i'}$, we say that they have the same type $\tau(i) = \tau(i')$. If $\tau(i) = \tau(i') := \tau$, we write $w_{ij} = w_{i'j} := w_{\tau j}$

[Next verbatim source excerpt]

symmetric structure. Suppose users come in K types, where a user of type k shares a utility vector. Then, consider $S_{\text{symm}} = \{p : p_i = p_{i'} \text{ if } w_i = w_{i'}\} \subseteq \Delta_m$ $n-1$, the set of policies where users of the same type are given the same recommendation probabilities. Note that, in the extreme case, each user is their own type. Furthermore, let p_k denote the recommendation policy for type k , for $p \in S_{\text{symm}}$.

[Next verbatim source excerpt]

We first define a piece of convenient notation: if two users i, i' share the same utility vector $w_i = w_{i'}$, we say that they have the same type $\tau(i) = \tau(i')$. If $\tau(i) = \tau(i') := \tau$, we write $w_{ij} = w_{i'j} := w_{\tau j}$

Expanded PaperInterface specification

```
∀ {m n K : ℕ} [inst : NeZero m] [inst_1 : NeZero n] [NeZero K]
(S : GCG24UserItemFairness.RecommendationModel.SymmetricData m n K) (hTypes : Function.Surjective S.types.toType),
GCG24UserItemFairness.assumption_positive_recommendation_utilities S.model →
GCG24UserItemFairness.assumption_proposition2_utility_rows_determine_types S →
  ∀ (psrc : GCG24UserItemFairness.Policy m n) (ell : ℝ)
    (hsource :
      (GCG24UserItemFairness.RecommendationModel.UserSymmetricEqualityLP.ofSymmetricData S).ActiveBasis psrc ell),
  (∀ (p : GCG24UserItemFairness.Policy m n),
    S.model.UtilityRowSymmetric p ↔
      ∀ (c : GCG24UserItemFairness.UserTypeAssignment.TypeSymmetryLinearConstraintIndex m n),
        GCG24UserItemFairness.RecommendationModel.policyLinearConstraintHolds
          (GCG24UserItemFairness.UserTypeAssignment.typeSymmetryLinearCoefficient S.types c) 0
          GCG24UserItemFairness.RecommendationModel.PolicyLinearComparison.eq p) ∧
    (∃ (psym, S.model.UtilityRowSymmetric psym ∧ S.model.IsOptimalAtLevel 1 psym) ∧
      (GCG24UserItemFairness.UserTypeAssignment.descendTypePolicy S.types
        (EconCSLib.Policy.FiberRepresentatives.ofSurjective hTypes) psrc).ActivePairsBound ∧
      (ell = (S.canonicalReductionOfSurjective hTypes).reduced.optimalItemFairness →
        (GCG24UserItemFairness.UserTypeAssignment.descendTypePolicy S.types
          (EconCSLib.Policy.FiberRepresentatives.ofSurjective hTypes) psrc).SharedItemsBound))
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.proposition2_symmetric_optimum_exists

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

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Review row 11

Source locator: cited publication, lines 126-132

Verbatim source input

utility from
P
P
P
using the platform is $\sum_j p_{ij} w_{ij}$; item j 's expected utility is $\sum_i p_{ij} w_{ij}$, where $\sum_i p_{ij} = 1$ for each user i .
Fairness desiderata. We suppose that the platform uses as a benchmark, for each user or item,

Expanded PaperInterface specification

$\forall \{m, n : \mathbb{N}\} (W : \text{GCG24UserItemFairness.RecommendationModel } m \ n) (p : \text{GCG24UserItemFairness.Policy } m \ n)$
 $(j : \text{GCG24UserItemFairness.Item } n),$
 $\text{GCG24UserItemFairness.ProofBridge.rawItemUtility } W \ p \ j = \sum u, W.\text{utility } u \ j * ((p \ u) \ j).\text{toReal}$

Lean proof endpoint (built separately)

`GCG24UserItemFairness.PaperInterface.rawItemUtility_realizes_spec`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 126-132

Verbatim source input

utility from
P
P
P
using the platform is $\sum_j p_{ij} w_{ij}$; item j 's expected utility is $\sum_i p_{ij} w_{ij}$, where $\sum_j p_{ij} = 1$ for each user i .
Fairness desiderata. We suppose that the platform uses as a benchmark, for each user or item,

Expanded PaperInterface specification

$\forall \{m, n : \mathbb{N}\} (W : \text{GCG24UserItemFairness.RecommendationModel } m \ n) (p : \text{GCG24UserItemFairness.Policy } m \ n)$
 $(u : \text{GCG24UserItemFairness.User } m),$
 $\text{GCG24UserItemFairness.ProofBridge.rawUserUtility } W \ p \ u = \text{EconCSLib.Policy.agentScore } p \ W.\text{utility } u$

Lean proof endpoint (built separately)

`GCG24UserItemFairness.PaperInterface.rawUserUtility_realizes_spec`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 116-120

Verbatim source input

Users, items, and utilities. There is a finite population of m users and n items. Let $w_{ij} > 0$ be the utility of recommending item j to user i ; this could represent a click-through rate or purchase probability. We suppose that user-item utilities are symmetric: each of the user i and item j receives utility w_{ij} from being recommended item j . Let Δ_{n-1} denote the simplex in $\mathbb{R}^n$. The platform's task is to choose a recommendation policy

Expanded PaperInterface specification

```
∀ {m n : ℕ} (W : GCG24UserItemFairness.RecommendationModel m n) (u : GCG24UserItemFairness.User m)
  (j : GCG24UserItemFairness.Item n), GCG24UserItemFairness.ProofBridge.recommendationUtility W u j = W.utility u j
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.recommendationUtility_realizes_spec

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 152-174

Verbatim source input

```
fairness desiderata as follows. At the extremes, the platform could choose a maximally fair solution
*
for one side, ignoring the other. Denote the optimal user fair utility as  $U_{\min}$ 
 $(w) := \max_p U_{\min}(p, w)$ 
(achieved by giving each user their favorite item deterministically), and the optimal item fair utility
*
as  $I_{\min}$ 
 $(w) := \max_p I_{\min}(p, w)$ .
Finally, we cast the two-sided fair optimization – for the optimal  $\gamma$ -constrained user fair solution – as
*
 $U_{\min}$ 
 $(\gamma, w) = \max_p U_{\min}(p, w)$ 
 $p$ 

(1)

*
subject to  $I_{\min}(p, w) \geq \gamma I_{\min}(w)$ ,
(w),

i.e., we maximize the minimum normalized user utility, subject to the minimum normalized item
utility being at least a fraction  $\gamma$  of the optimal item fair solution.
Research questions: price of fairness and misestimation. We can now define the price of item
```

Expanded PaperInterface specification

```
∀ {m n : ℕ} [inst : NeZero m] [inst_1 : NeZero n] (W : GCG24UserItemFairness.RecommendationModel m n) (γ : ℝ)
  (p : GCG24UserItemFairness.Policy m n),
  GCG24UserItemFairness.ProofBridge.solvesProblemOne W γ p ↔ W.IsOptimalAtLevel γ p
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.solvesProblemOne_iff

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 15

Source locator: cited publication, lines 1742-1745

Verbatim source input

Lemma 1. Since $w_{ij} > 0$ for all j , $I_{\min}$
> 0.
*
= $\max_p I_{\min}(\rho) > 0$.

Expanded PaperInterface specification

$\forall \{m\ n : \mathbb{N}\} \text{ [NeZero } m] \text{ [inst : NeZero } n] \text{ (} W : \text{GCG24UserItemFairness.RecommendationModel } m\ n),$
 $W.\text{Positive} \rightarrow 0 < W.\text{optimalItemFairness}$

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_c_lemmal_item_fairness_positive

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 3071-3078

Verbatim source input

Lemma 10. If $\alpha \leq 1/2$, then $t(\alpha) \leq (n + 1)/2$.
Now, for $1 \leq t \leq n$, define $A(t) = \{\alpha : t(\alpha) = t\}$. Since $t(\alpha)$ is weakly increasing in α by Lemma 7, $A(t)$ is an interval in $(0, 1)$, and $[A(1), A(2), \dots, A(n)]$ forms a consecutive partition of $(0, 1)$. We
* will now show that $I_{\min}$
(α) is increasing on each of the first $\lfloor (n + 1)/2 \rfloor$ intervals, which by Lemma
10 contain all $\alpha \in (0, 1/2]$.
*

Expanded PaperInterface specification

```
∀ {n : ℕ} [inst : NeZero n] {alpha : ℝ} {v : GCG24UserItemFairness.Item n → ℝ} (halpha0 : 0 < alpha)
(halphal : alpha < 1),
alpha ≤ 1 / 2 →
  ∀ (hpos : ∀ (j : GCG24UserItemFairness.Item n), 0 < v j),
    GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v →
      have t :=
        (GCG24UserItemFairness.OpposingTypes.problem6FirstClosedPolicy alpha v halpha0 halphal
         hpos).lastActiveTypeZero;
      t ≤ ↑(GCG24UserItemFairness.OpposingTypes.reverseItem t)
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_d_lemmal0_active_optimizer_pivot_at_or_before_midpoint

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 17

Source locator: cited publication, lines 3079-3100

Verbatim source input

Lemma 11. $I_{\min}$
(α) is increasing on $A(t)$ for $t \leq (n+1)/2$.
*
Finally, we show that this implies $I_{\min}$
(α) is increasing on $0 < \alpha \leq 1/2$. Notice that $I_j(x, y, \alpha)$
is continuous in α for each fixed x, y , so $I_{\min}(x, y, \alpha) = \min_j I_j(x, y, \alpha)$ is continuous in α for
*
each fixed x, y . Recall that $I_{\min}$
(α) = $\max_{x, y \in \Delta^{n-1}} I_{\min}(x, y, \alpha)$. As the supremum of a set of
*
continuous functions on a compact set must itself be continuous, this means that $I_{\min}$
(α) must be
*
continuous in α , since $\Delta^{n-1} \times \Delta^{n-1}$ is compact. Since $I_{\min}(\alpha)$ is continuous and is increasing on
*
each interval $\{A(t) \mid t \leq (n+1)/2\}$ by Lemma 11, $I_{\min}$
(α) must be increasing on $t \leq (n+1)/2$ $A(t)$. By
S
*
Lemma 10, $t \leq (n+1)/2$ $A(t) \supset (0, 1/2]$. So $I_{\min}(\alpha)$ is increasing on $(0, 1/2]$.
Now, we show Lemmas 10 and 11.

Expanded PaperInterface specification

```
∀ {n : ℕ} [inst : NeZero n] {alpha alpha' : ℝ} {v : GCG24UserItemFairness.Item n → ℝ} (halpha0 : 0 < alpha)
(halpha1 : alpha < 1) (halpha0' : 0 < alpha') (halpha1' : alpha' < 1),
alpha ≤ alpha' →
  ∀ (hpos : ∀ (j : GCG24UserItemFairness.Item n), 0 < v j),
    GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v →
      (GCG24UserItemFairness.OpposingTypes.problem6FirstClosedPolicy alpha v halpha0 halpha1
        hpos).lastActiveTypeZero =
        (GCG24UserItemFairness.OpposingTypes.problem6FirstClosedPolicy alpha' v halpha0' halpha1'
          hpos).lastActiveTypeZero →
        (have t :=
          (GCG24UserItemFairness.OpposingTypes.problem6FirstClosedPolicy alpha v halpha0 halpha1
            hpos).lastActiveTypeZero;
          ↑t ≤ ↑(GCG24UserItemFairness.OpposingTypes.reverseItem t)) →
          (GCG24UserItemFairness.OpposingTypes.twoTypeReducedModel alpha v).optimalItemFairness ≤
          (GCG24UserItemFairness.OpposingTypes.twoTypeReducedModel alpha' v).optimalItemFairness
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_d_lemma11_optimal_item_fairness_mono_on_active_pivot_interval

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 18

Source locator: cited publication, lines 2098-2182

Verbatim source input

```
Lemma 3. For all  $0 < \alpha < 1$ ,  $U_{\min}(\alpha) = 1$ .
*
*
F
 $(U_{\min} - U_{\min}(1, \alpha)) / U$ 
By Lemma 3,  $U_{\min}(\alpha) := U_{\min}$ 
is constant in  $\alpha$ , so  $\pi U_{\min}$  and it is
 $|I(\alpha)| =$ 
*
enough to show  $U_{\min}(1, \alpha)$  increases in  $\alpha$ .
*

*

Recall the definition of  $S_{\text{symm}}$  :
 $\{p : p_i = p_i' \text{ if } w_i = w_i'\}$ .
By Proposition 2, we know that  $S_{\text{symm}}$  satisfies conditions (i) and (ii) on  $S$  for Proposition 1.
Thus by Proposition 1, it is sufficient to find  $\phi$  such that
 $\phi \in \arg$ 

max

 $p \in S_{\text{symm}}^{n-1, \lambda}$ 

 $\lambda$ 

subject to  $I_j(p) = \lambda$ 

 $\forall j$ ,

and show that this  $\phi$  is unique, where the linear program is derived from the proof of Proposition 1.
Since each user in our population only has one of two possible value vectors,  $w_i = (v_1, \dots, v_n)$ 
or  $w_i = (v_n, \dots, v_1)$ , we can identify  $S_{\text{symm}}$  with  $\Delta_{2n-1}$ , and identify  $p \in S_{\text{symm}}$  with a pair  $x, y$ .
Expanding the expression for  $I_j(p)$  and making explicit the dependence on  $\alpha$ , we see that

$$I_j(p, \alpha) = \sum_i p_{ij} w_{ij}(\alpha) = \sum_i p_i \sum_j w_{ij}(\alpha)$$


$$= \sum_i \tau(i) p_i v_j + \sum_i \tau(i) p_i v_{n-j+1}$$


$$= \sum_i \tau(i) p_i v_j + \sum_i \tau(i) p_i v_{n-j+1}$$


$$= \sum_j \alpha x_j + (1 - \alpha) y_j$$

where
 $q_j(\alpha) := \alpha v_j + (1 - \alpha) v_{n-j+1}$ 
Note that for all  $j$  and  $0 < \alpha < 1$ , we have  $0 < q_j$ ,  $1 - q_j < 1$ . We can thus simplify
 $\phi = \arg$ 

max

 $x, y \in \Delta_{2n-1}, \lambda$ 

 $\lambda$ 
(6)

subject to  $q_j(\alpha) x_j + (1 - q_j(\alpha)) y_j = \lambda$ 

 $\forall j$ ,

First, we will show uniqueness and find a closed form to Problem 6 in Lemmas 4 and 5. Then, we
*
will use this closed forms to show that  $U_{\min}(1, \alpha)$  is increasing for  $\alpha < 1/2$ . By symmetry, this
*
implies that  $U_{\min}(1, \alpha)$  must be decreasing for  $\alpha > 1/2$  and we will be done.
```

Expanded PaperInterface specification

$\forall \{m\ n : \mathbb{N}\} \text{ [inst : NeZero } m] \text{ [inst_1 : NeZero } n] \text{ (W : GCG24UserItemFairness.RecommendationModel } m\ n),$
W.Nonnegative $\rightarrow$ W.RowHasPositiveItem $\rightarrow$ W.optimalUserFairnessAtLevel 0 = 1

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_d_lemma3_unconstrained_baseline

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 19

Source locator: cited publication, lines 2183-2188

Verbatim source input

Lemma 4. Problem 6 has a unique solution (x, y, λ) . Moreover, the solution is sparse: there is some $1 \leq t \leq n$ such that for $j > t$, $x_j = 0$, and for $j < t$, $y_j = 0$.
Thus, to solve our original problem (Problem 1) we merely need to evaluate the user fairness $U_{\min}$ of x, y solving Problem 6 and show that this is increasing. We will do this in the remainder of the proof.
For each α , let x, y be the solution to the corresponding Problem 6, and define
 $t(\alpha) := \max\{j : x_j(\alpha) > 0\}$.

Expanded PaperInterface specification

```
∀ {n : ℕ} [NeZero n] {alpha : ℝ} {v : GCG24UserItemFairness.Item n → ℝ},
  0 < alpha →
  alpha < 1 →
  (∀ (j : GCG24UserItemFairness.Item n), 0 < v j) →
  GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v →
  ∃ t pstar ellstar,
    (∀ (j : GCG24UserItemFairness.Item n),
      GCG24UserItemFairness.OpposingTypes.pairShare alpha v j * ((pstar 0) j).toReal +
      (1 - GCG24UserItemFairness.OpposingTypes.pairShare alpha v j) * ((pstar 1) j).toReal =
      ellstar) ∧
    (∀ (p : GCG24UserItemFairness.TypePolicy 2 n) (ell : ℝ),
      GCG24UserItemFairness.OpposingTypes.problem6LPFeasible alpha v p ell → ell ≤ ellstar) ∧
    (∀ {j : GCG24UserItemFairness.Item n}, ↑t < ↑j → ((pstar 0) j).toReal = 0) ∧
    (∀ {j : GCG24UserItemFairness.Item n}, ↑j < ↑t → ((pstar 1) j).toReal = 0) ∧
    ∀ (p : GCG24UserItemFairness.TypePolicy 2 n) (ell : ℝ),
      (∀ (j : GCG24UserItemFairness.Item n),
        GCG24UserItemFairness.OpposingTypes.pairShare alpha v j * ((p 0) j).toReal +
        (1 - GCG24UserItemFairness.OpposingTypes.pairShare alpha v j) * ((p 1) j).toReal =
        ell) →
      (∀ (p' : GCG24UserItemFairness.TypePolicy 2 n) (ell' : ℝ),
        GCG24UserItemFairness.OpposingTypes.problem6LPFeasible alpha v p' ell' → ell' ≤ ell) →
      p = pstar ∧ ell = ellstar
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_d_lemma4_problem6_unique_sparse_solution

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 20

Source locator: cited publication, lines 2190-2256

Verbatim source input

Lemma 5. Let $0 < \alpha < 1$, $t := t(\alpha)$. Define

```
X 1
Lt =
  ,
  q
j< t j

Rt =
  7
  1
  ;
  1 - qj
j> t

X
```

While there is a single optimal solution x, y to Problem 6 for each α , there may be two t that satisfy Lemma 4. In particular, this occurs when for each j , $x_j = 0$ or $y_j = 0$. By contrast, $t(\alpha)$ is unique.

28

[form-feed in source extraction]
Then

```
1
  ,
  1 + qt Lt + (1 - qt )Rt
  *
  Imin
  =

and

[U+F8F1 delimiter glyph] I*
min
[U+F8F4 delimiter glyph]
j< t
[U+F8F2 delimiter glyph] qj ,
Lt
xj = 1 - 1+q L +(1-q
  , j=t,
  t t
  t )Rt
[U+F8F4 delimiter glyph]
[U+F8F3 delimiter glyph]
0,
j> t

[U+F8F1 delimiter glyph]
[U+F8F4 delimiter glyph]
[U+F8F2 delimiter glyph]0,
Rt
  ,
  yj = 1 - 1+qt Lt +(1-q
  t )Rt
  *
  [U+F8F4 delimiter glyph]
  [U+F8F3 delimiter glyph] Imin ,
  1-qj

j< t
j=t.
j> t
```

Note that for $t \neq t(\alpha)$, x, y above may not be valid probability vectors.

Expanded PaperInterface specification

```
∀ {n : ℕ} [inst : NeZero n] {alpha : ℝ} {v : GCG24UserItemFairness.Item n → ℝ},
  0 < alpha →
  alpha < 1 →
  (∀ (j : GCG24UserItemFairness.Item n), 0 < v j) →
  GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v →
  ∃ pstar ellstar,
  (∀ (j : GCG24UserItemFairness.Item n),
    GCG24UserItemFairness.OpposingTypes.pairShare alpha v j * ((pstar 0) j).toReal +
    (1 - GCG24UserItemFairness.OpposingTypes.pairShare alpha v j) * ((pstar 1) j).toReal =
    ellstar) ∧
  (∀ (p : GCG24UserItemFairness.TypePolicy 2 n) (ell : ℝ),
    GCG24UserItemFairness.OpposingTypes.problem6LPFeasible alpha v p ell → ell ≤ ellstar) ∧
  (∀ (p : GCG24UserItemFairness.TypePolicy 2 n) (ell : ℝ),
    (∀ (j : GCG24UserItemFairness.Item n),
      GCG24UserItemFairness.OpposingTypes.pairShare alpha v j * ((p 0) j).toReal +
      (1 - GCG24UserItemFairness.OpposingTypes.pairShare alpha v j) * ((p 1) j).toReal =
```

```

    ell) →
  (∀ (ρ' : GCG24UserItemFairness.TypePolicy 2 n) (ell' : ℝ),
    GCG24UserItemFairness.OpposingTypes.problem6LPFeasible alpha v ρ' ell' → ell' ≤ ell) →
    ρ = pstar ∧ ell = ellstar) ∧
have t := pstar.lastActiveTypeZero;
ellstar = GCG24UserItemFairness.OpposingTypes.problem6ClosedValue alpha v t ∧
(∀ {j : GCG24UserItemFairness.Item n},
  ↑j < ↑t →
    ((pstar 0) j).toReal = ellstar / GCG24UserItemFairness.OpposingTypes.pairShare alpha v j) ∧
  ((pstar 0) t).toReal =
    1 - ellstar * GCG24UserItemFairness.OpposingTypes.problem6LeftSum alpha v t ∧
  (∀ {j : GCG24UserItemFairness.Item n}, ↑t < ↑j → ((pstar 0) j).toReal = 0) ∧
  (∀ {j : GCG24UserItemFairness.Item n}, ↑j < ↑t → ((pstar 1) j).toReal = 0) ∧
  ((pstar 1) t).toReal =
    1 - ellstar * GCG24UserItemFairness.OpposingTypes.problem6RightSum alpha v t ∧
  (∀ {j : GCG24UserItemFairness.Item n},
    ↑t < ↑j →
      ((pstar 1) j).toReal =
        ellstar / (1 - GCG24UserItemFairness.OpposingTypes.pairShare alpha v j)

```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_d_lemma5_problem6_active_optimizer_closed_form

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 21

Source locator: cited publication, lines 2257-2313

Verbatim source input

Lemma 6. For $0 < \alpha < 1/2$, if ρ^* is the optimal recommendation policy, then $U_i(\rho^*, \alpha) \geq U_i'(\rho^*, \alpha)$ for $\tau(i) = 1$, $\tau(i') = 2$.

*
By Lemma 6, $U_{\min}(\alpha) = U_i(\alpha)$ for i with $\tau(i) = 2$. Then
*
 $U_{\min}(1, \alpha) = U_i(\alpha)$
n
X
1
yj vn-j+1
maxj vj j=1
[U+F8EB delimiter glyph]
[U+F8F6 delimiter glyph]
X
1
[U+F8ED delimiter glyph]yt(α) vn-t(α)+1 +
yj vn-j+1 [U+F8F8 delimiter glyph]
=
maxj vj
j>t(α)
[U+F8EB delimiter glyph][U+F8EB delimiter glyph]
[U+F8F6 delimiter glyph]
[U+F8F6 delimiter glyph]
X I * (α)
X I * (α)
1
min
min
[U+F8ED delimiter glyph][U+F8ED delimiter glyph]1 -
[U+F8F8 delimiter glyph] vn-t(α)+1 +
=
vn-j+1 [U+F8F8 delimiter glyph]
maxj vj
1 - qj (α)
1 - qj (α)
j>t(α)
j>t(α)
[U+F8EB delimiter glyph]
[U+F8F6 delimiter glyph]
X I * (α)
1
min
[U+F8ED delimiter glyph]vn-t(α)+1 +
=
(vn-j+1 - vn-t(α)+1) [U+F8F8 delimiter glyph]
maxj vj
1 - qj (α)
=
j>t(α)
*
The following results will enable us to conclude that $U_{\min}(1, \alpha)$ is increasing for $\alpha < 1/2$.

Expanded PaperInterface specification

$\forall \{n : \mathbb{N}\} [\text{inst} : \text{NeZero } n] \{ \alpha : \mathbb{R} \} \{ v : \text{GCG24UserItemFairness.Item } n \rightarrow \mathbb{R} \} (\text{halpha0} : 0 < \alpha)$
 (halpa_half : $\alpha < 1 / 2$) (hpos : $\forall (j : \text{GCG24UserItemFairness.Item } n), 0 < v j$),
 GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v $\rightarrow$
 (GCG24UserItemFairness.OpposingTypes.twoTypeReducedModel alpha v).normalizedTypeUtility
 (GCG24UserItemFairness.OpposingTypes.problem6FirstClosedPolicy alpha v halpha0 ... hpos) $1 \leq$
 (GCG24UserItemFairness.OpposingTypes.twoTypeReducedModel alpha v).normalizedTypeUtility
 (GCG24UserItemFairness.OpposingTypes.problem6FirstClosedPolicy alpha v halpha0 ... hpos) 0

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_d_lemma6_optimal_policy_type_utility_order

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 22

Source locator: cited publication, lines 2315-2316

Verbatim source input

Lemma 7. $t(\alpha)$ is weakly increasing.

*

Expanded PaperInterface specification

```
∀ {n : ℕ} [inst : NeZero n] {alpha alpha' : ℝ} {v : GCG24UserItemFairness.Item n → ℝ} (halpha0 : 0 < alpha)
(halpha1 : alpha < 1) (halpha0' : 0 < alpha') (halpha1' : alpha' < 1),
alpha ≤ alpha' →
  ∀ (hpos : ∀ (j : GCG24UserItemFairness.Item n), 0 < v j),
    GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v →
      ↑(GCG24UserItemFairness.OpposingTypes.problem6FirstClosedPolicy alpha v halpha0 halpha1
        hpos).lastActiveTypeZero ≤
        ↑(GCG24UserItemFairness.OpposingTypes.problem6FirstClosedPolicy alpha' v halpha0' halpha1'
          hpos).lastActiveTypeZero
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_d_lemma7_active_optimizer_pivot_monotonicity

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 23

Source locator: cited publication, lines 2317-2318

Verbatim source input

Lemma 8. $I_{\min}$
(α) is increasing for $\alpha \leq 1/2$.

Expanded PaperInterface specification

```
∀ {n : ℕ} [inst : NeZero n] {alpha alpha' : ℝ} {v : GCG24UserItemFairness.Item n → ℝ},
  0 < alpha →
  alpha < 1 →
  0 < alpha' →
  alpha' < 1 →
  alpha ≤ alpha' →
  alpha ≤ 1 / 2 →
  alpha' ≤ 1 / 2 →
  (∀ (j : GCG24UserItemFairness.Item n), 0 < v j) →
  GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v →
  (GCG24UserItemFairness.OpposingTypes.twoTypeReducedModel alpha v).optimalItemFairness ≤
  (GCG24UserItemFairness.OpposingTypes.twoTypeReducedModel alpha' v).optimalItemFairness
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_d_lemma8_selected_pivot_stitching

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 24

Source locator: cited publication, lines 2320-2339

Verbatim source input

Lemma 9. $q_j(\alpha)$ strictly increases as α increases and strictly decreases as j increases.
* Since $t(\alpha)$ is increasing and v_j decreases in j , $v_n - t(\alpha) + 1$ is increasing in α . Also, $I_{\min}(\alpha)$ is
I*

(α)

min
increasing for $\alpha < 1/2$, and $q_j(\alpha)$ is increasing, so $1 - q$
is increasing. For $j > t(\alpha)$, $v_n - j + 1 -$
 $j(\alpha)$

*
 $v_n - t(\alpha) + 1 > 0$. Thus, $U_{\min}(1, \alpha)$ is increasing.

D.2

Supporting lemma proofs

We will first prove Lemma 9, as it is a simple result that we will use extensively.

Expanded PaperInterface specification

```
∀ {n : ℕ} {alpha alpha' : ℝ} {v : GCG24UserItemFairness.Item n → ℝ} {i j : GCG24UserItemFairness.Item n},
  0 < alpha →
  alpha < 1 →
  0 < alpha' →
  alpha' < 1 →
  alpha < alpha' →
  (∀ (j : GCG24UserItemFairness.Item n), 0 < v j) →
  GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v →
  ↑i < ↑j →
  GCG24UserItemFairness.OpposingTypes.pairShare alpha v j <
    GCG24UserItemFairness.OpposingTypes.pairShare alpha' v j ∧
  GCG24UserItemFairness.OpposingTypes.pairShare alpha v j <
    GCG24UserItemFairness.OpposingTypes.pairShare alpha v i
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_d_lemma9_pair_share_algebra

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 25

Source locator: cited publication, lines 3379-3402

Verbatim source input

Lemma 12. If a policy $\rho = (x, y, z) \in \text{Ssymm}$ solves $U_{\min}(\rho) = U_{\min}(1)$, then there is some policy

*
 $\rho' = x', z' \in S'$ that solves $U_{\min}(\rho') = U_{\min}(1)$.

Applying the framework in Proposition 1 to this proof with $S = S'$ and using $\hat{w}$ as the user and item values, we reduce the problem to finding
 $\phi = \arg$

$\max$

$x, z \in \Delta^{2n-1}, \lambda$

λ
(11)

subject to $I_j(x, z) = \lambda$

$\forall j,$

and showing that this ϕ is unique.

First, we use an analogue of the sparsity result Proposition 2 to show that the optimal x, z has the following sparse structure.

Expanded PaperInterface specification

```
∀ {n : ℕ} [inst : NeZero n] {beta : ℝ} {v : GCG24UserItemFairness.Item n → ℝ},
  (∀ (j : GCG24UserItemFairness.Item n), 0 < v j) →
  ∀ {ρ : GCG24UserItemFairness.TypePolicy 3 n},
    (GCG24UserItemFairness.OpposingTypes.theorem4EstimatedReducedModel beta v).IsOptimalAtLevel 1 ρ →
    (GCG24UserItemFairness.OpposingTypes.theorem4EstimatedReducedModel beta v).IsOptimalAtLevel 1
      (GCG24UserItemFairness.OpposingTypes.theorem4SymmetrizedPolicy ρ) ∧
      (GCG24UserItemFairness.OpposingTypes.Theorem4MirrorSymmetricPolicy
        (GCG24UserItemFairness.OpposingTypes.theorem4SymmetrizedPolicy ρ))
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_e_lemmal2_symmetrized_policy

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 26

Source locator: cited publication, lines 3403-3412

Verbatim source input

Lemma 13. Let x, z be an optimal basic feasible solution to the linear program Problem 11. There

,
,
,
,

is some $j \leq n+1$
2 such that for all $j > j$, $x_j = 0$ and for all $j < j$, $z_j = z_{n-j+1} = 0$. Let the
pivot index t denote the minimum such j .
This sparse structure implies uniqueness.

[Next verbatim source excerpt]

$S' = \{x, y, z \in S_{\text{symm}} : x_j = y_{n-j+1}, z_j = z_{n-j+1} \text{ for all } j\}$.
Since y is completely determined by x , we can identify S' with Δ^{2n-1} and can identify $x, y, z \in S'$
simply by x, z .

*
Lemma 12. If a policy $\rho = (x, y, z) \in S_{\text{symm}}$ solves $U_{\min}(\rho) = U_{\min}$
(1), then there is some policy

*
 $\rho' = x', z' \in S'$ that solves $U_{\min}(\rho') = U_{\min}$
(1).

Applying the framework in Proposition 1 to this proof with $S = S'$ and using $\hat{w}$ as the user and item
values, we reduce the problem to finding
 $\phi = \arg$

max

$x, z \in \Delta^{2n-1}, \lambda$

λ
(11)

subject to $I_j(x, z) = \lambda$

$\forall j$,

and showing that this ϕ is unique.

[Next verbatim source excerpt]

Lemma 17. Let (x, z) be an optimal solution to Problem 11. For $j > n+1$
2, $x_j = 0$.

In particular, if we let $h = \lfloor n+1$
2 $\rfloor$ indicate the index halfway through the set of items, this means that
we can simplify the problem above to
 $\arg \max$

$x_1, \dots, x_h,$

$z_1, \dots, z_h,$

λ

h

$z_1, \dots, z_h,$
 λ

subject to $I_j(x, z) = \lambda, 1 \leq j \leq h$

x

x

x

$x_j =$

$z_j +$

$z_{n-j+1} = 1$

$j \leq h$

$j \leq h$

(12)

$h < j \leq n$

$x_j, z_j \geq 0$.

Now this is a linear program with $2h+1$ variables and $h+2$ constraints, so by an argument analogous
to Problem 12, we find that every basic feasible solution $x_1:h, z_1:h$ must share $h-1$ zeros. This means
that there is a single index t such that $x_t > 0, z_t > 0$; if there were more than one, by the pigeonhole
principle there must be some $t' \leq h$ such that $x_{t'} = z_{t'} = x_{n-t'+1} = 0$, and $I_{t'}(x, z) = 0$. Then

*
 $I_{\min}(x, z) = 0$; however, by Lemma 1, $I_{\min}(x, z) = I_{\min}$
 > 0 , so we have a contradiction.

Expanded PaperInterface specification

$\forall \{n : \mathbb{N}\} [\text{inst} : \text{NeZero } n] \{\text{beta} : \mathbb{R}\} \{v : \text{GCG24UserItemFairness.Item } n \rightarrow \mathbb{R}\} \{\rho : \text{GCG24UserItemFairness.TypePolicy } 3 \ n\}$
 $\{\text{ell} : \mathbb{R}\},$
 $2 < n \rightarrow$
 $(\uparrow n)^{-1} < \text{beta} \rightarrow$

```

beta < 1 / 2 →
  (∀ (j : GCG24UserItemFairness.Item n), 0 < v j) →
    GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v →
      GCG24UserItemFairness.OpposingTypes.Theorem4MirrorSymmetricPolicy p →
        (∀ (j : GCG24UserItemFairness.Item n),
          GCG24UserItemFairness.OpposingTypes.theorem4Problem11PolicyItemValue beta v p j = ell) →
          (∀ (p' : GCG24UserItemFairness.TypePolicy 3 n) (ell' : ℝ),
            GCG24UserItemFairness.OpposingTypes.Theorem4MirrorSymmetricPolicy p' →
              (∀ (j : GCG24UserItemFairness.Item n),
                GCG24UserItemFairness.OpposingTypes.theorem4Problem11PolicyItemValue beta v p' j = ell') →
                ell' ≤ ell) →
            GCG24UserItemFairness.ProofBridge.problem11BasicFeasible beta v p ell →
              have t := GCG24UserItemFairness.OpposingTypes.theorem4Problem11LastActiveTypeZero p;
              †t ≤ †(GCG24UserItemFairness.OpposingTypes.reverseItem t) ∧
                (∀ (j : GCG24UserItemFairness.Item n), †t < †j → (p 0) j = 0) ∧
                (∀ (j : GCG24UserItemFairness.Item n),
                  †j < †t → (p 2) j = 0 ∧ (p 2) (GCG24UserItemFairness.OpposingTypes.reverseItem j) = 0) ∧
                  ∀ (s : GCG24UserItemFairness.Item n),
                    (∀ (j : GCG24UserItemFairness.Item n), †s < †j → (p 0) j = 0) →
                    (∀ (j : GCG24UserItemFairness.Item n),
                      †j < †s →
                        (p 2) j = 0 ∧ (p 2) (GCG24UserItemFairness.OpposingTypes.reverseItem j) = 0) →
                      †t ≤ †s
                )

```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_e_lemma13_every_optimal_basic_solution_has_pivot_support

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 27

Source locator: cited publication, lines 3413-3414

Verbatim source input

Lemma 14. Problem 11 has a unique optimal solution.
We can now describe what this solution (x, z) looks like.

Expanded PaperInterface specification

```
∀ {n : ℕ} [NeZero n] {beta : ℝ} {v : GCG24UserItemFairness.Item n → ℝ},
  2 < n →
  (1/n)-1 < beta →
  beta < 1 / 2 →
  (∀ (j : GCG24UserItemFairness.Item n), 0 < v j) →
  GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v →
  ∃ pstar ellstar,
    GCG24UserItemFairness.OpposingTypes.Theorem4MirrorSymmetricPolicy pstar ∧
    (∀ (j : GCG24UserItemFairness.Item n),
      GCG24UserItemFairness.OpposingTypes.theorem4Problem11PolicyItemValue beta v pstar j = ellstar) ∧
    (∀ (p : GCG24UserItemFairness.TypePolicy 3 n) (ell : ℝ),
      GCG24UserItemFairness.OpposingTypes.Theorem4MirrorSymmetricPolicy p →
      GCG24UserItemFairness.OpposingTypes.theorem4Problem11LPFeasible beta v p ell → ell ≤ ellstar) ∧
    ∀ (p : GCG24UserItemFairness.TypePolicy 3 n) (ell : ℝ),
      GCG24UserItemFairness.OpposingTypes.Theorem4MirrorSymmetricPolicy p →
      GCG24UserItemFairness.OpposingTypes.theorem4Problem11LPFeasible beta v p ell →
      (∀ (p' : GCG24UserItemFairness.TypePolicy 3 n) (ell' : ℝ),
        GCG24UserItemFairness.OpposingTypes.Theorem4MirrorSymmetricPolicy p' →
        GCG24UserItemFairness.OpposingTypes.theorem4Problem11LPFeasible beta v p' ell' →
        ell' ≤ ell) →
      p = pstar ∧ ell = ellstar
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_e_lemma14_problem11_has_unique_optimal_solution

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 28

Source locator: cited publication, lines 3415-3472

Verbatim source input

```
Lemma 15. Let (x, z) be the optimal solution to Problem 12, and let t be the pivot element of (x, z);
suppose that t  $\neq$  n+1
2 . Define
x 1
Lt :=
.
q
j<t j
Then
 $\lambda =$ 
[U+F8F1 delimiter glyph]
[U+F8F4 delimiter glyph]
[U+F8F2 delimiter glyph]0,[U+0010 control character]

zj = 12
[U+F8F4 delimiter glyph]
[U+F8F3 delimiter glyph]

2 $\beta$ qt + 1/2(1 - 2 $\beta$ )
'
1 + qt Lt + 1/2(n - 2t)
j<t

 $\lambda$ 
1 - (n - 2t) 1-2 $\beta$ 
[U+0011 control character]

,

 $\lambda$ 
1-2 $\beta$  ,

j  $\in$  {t, n - t + 1} .
t<j <n-t+1

Finally,
[U+F8F1 delimiter glyph]
 $\lambda$ 
[U+F8F4 delimiter glyph]
[U+F8F2 delimiter glyph] 2 $\beta$ qj ,
 $\lambda$ 
xj = 1 - 2 $\beta$ 
Lt ,
[U+F8F4 delimiter glyph]
[U+F8F3 delimiter glyph]0,
38

j<t
j=t.
j>t

[form-feed in source extraction]
If t = n+1
2 , then x remains the same, but zt = 1 and
2 $\beta$ qt + (1 - 2 $\beta$ )
 $\lambda =$ 
.
1 + qt Lt
```

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

Appendix E, Lemma 15 corrected target: for the equality-form LP solved in the lemma, the printed noncenter λ , x , and z
 $\leftrightarrow$ coordinate formulas hold; at the center pivot $z_t = 1$ and $\lambda = 1 / (1 + L_t)$.

Archival source anchor: cited publication, lines 3415-3472

Recorded basis Repository user approved the Appendix E Lemma 15 typo correction in docs/APPROVED_CORRECTED_TARGET_E15_2026-07-27.md.

Expanded PaperInterface specification

```
 $\forall \{n : \mathbb{N}\} [inst : \text{NeZero } n] \{beta : \mathbb{R}\} \{v : \text{GCG24UserItemFairness.Item } n \rightarrow \mathbb{R}\} \{p : \text{GCG24UserItemFairness.TypePolicy } 3 \ n\}$ 
{ell :  $\mathbb{R}$ },
2 < n  $\rightarrow$ 
(1n)-1 < beta  $\rightarrow$ 
beta < 1 / 2  $\rightarrow$ 
( $\forall (j : \text{GCG24UserItemFairness.Item } n), 0 < v \ j) \rightarrow$ 
GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v  $\rightarrow$ 
GCG24UserItemFairness.OpposingTypes.Theorem4MirrorSymmetricPolicy p  $\rightarrow$ 
( $\forall (j : \text{GCG24UserItemFairness.Item } n),$ 
GCG24UserItemFairness.OpposingTypes.theorem4Problem11PolicyItemValue beta v p j = ell)  $\rightarrow$ 
( $\forall (p' : \text{GCG24UserItemFairness.TypePolicy } 3 \ n) (ell' : \mathbb{R}),$ 
GCG24UserItemFairness.OpposingTypes.Theorem4MirrorSymmetricPolicy p'  $\rightarrow$ 
```

```

    GCG24UserItemFairness.OpposingTypes.theorem4Problem11LPFeasible beta v p' ell' → ell' ≤ ell) →
have t := GCG24UserItemFairness.OpposingTypes.theorem4Problem11LastActiveTypeZero p;
↑t ≤ ↑(GCG24UserItemFairness.OpposingTypes.reverseItem t) ∧
(∀ (j : GCG24UserItemFairness.Item n),
  ↑j < ↑t →
    ((p 0) j).toReal =
      ell / (2 * beta * GCG24UserItemFairness.OpposingTypes.pairShare (1 / 2) v j)) ∧
    ((p 0) t).toReal =
      1 -
        ell / (2 * beta) * GCG24UserItemFairness.OpposingTypes.paper_lemma15_problem11_leftSum v t +
    (∀ (j : GCG24UserItemFairness.Item n), ↑t < ↑j → ((p 0) j).toReal = 0) ∧
    (∀ (j : GCG24UserItemFairness.Item n),
      ↑j < ↑t →
        ((p 2) j).toReal = 0 ∧
        ((p 2) (GCG24UserItemFairness.OpposingTypes.reverseItem j)).toReal = 0) ∧
    (∀ (j : GCG24UserItemFairness.Item n),
      ↑t < ↑j →
        ↑j < ↑(GCG24UserItemFairness.OpposingTypes.reverseItem t) →
        ((p 2) j).toReal = ell / (1 - 2 * beta)) ∧
    (↑t < ↑(GCG24UserItemFairness.OpposingTypes.reverseItem t) ∧
      ell =
        (2 * beta * GCG24UserItemFairness.OpposingTypes.pairShare (1 / 2) v t +
          1 / 2 * (1 - 2 * beta)) /
        (1 +
          GCG24UserItemFairness.OpposingTypes.pairShare (1 / 2) v t *
          GCG24UserItemFairness.OpposingTypes.paper_lemma15_problem11_leftSum v t +
          1 / 2 * (↑n - 2 * (↑t + 1))) ∧
        ((p 2) t).toReal = 1 / 2 * (1 - (↑n - 2 * (↑t + 1)) * ell / (1 - 2 * beta)) ∧
        ((p 2) (GCG24UserItemFairness.OpposingTypes.reverseItem t)).toReal =
          1 / 2 * (1 - (↑n - 2 * (↑t + 1)) * ell / (1 - 2 * beta)) v
      ↑t = ↑(GCG24UserItemFairness.OpposingTypes.reverseItem t) ∧
      ((p 2) t).toReal = 1 ∧
      ell =
        1 / (1 + GCG24UserItemFairness.OpposingTypes.paper_lemma15_problem11_leftSum v t))

```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_e_lemma15_optimal_solution_full_closed_form

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 29

Source locator: cited publication, lines 3713-3735

Verbatim source input

Lemma 16. If $j < n+1$
2, then $q_j > 1/2$; if $j > 2$, then $q_j < 1/2$. If $j = 2$, $q_j = 1/2$.

Proof. If $j < n+1$
2, then $v_j > v_{n-j+1}$ and
 $q_j =$

$$\frac{v_j}{1+1}$$
$$= \frac{v_j}{2}$$
$$= \frac{v_{n-j+1}}{2}$$
$$= \frac{v_j + v_{n-j+1}}{2}$$

if $j < n+1$
2, then $v_j < v_{n-j+1}$ and a symmetric argument holds.
If $j = n+1$
2, then $v_j = v_{n-j+1}$ and thus $q_j = 1/2$.

Expanded PaperInterface specification

```
∀ {n : ℕ} {v : GCG24UserItemFairness.Item n → ℝ} (j : GCG24UserItemFairness.Item n),
  (∀ (j : GCG24UserItemFairness.Item n), 0 < v j) →
    GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v →
      (↑j < ↑(GCG24UserItemFairness.OpposingTypes.reverseItem j) →
        1 / 2 < GCG24UserItemFairness.OpposingTypes.pairShare (1 / 2) v j) ∧
      (↑(GCG24UserItemFairness.OpposingTypes.reverseItem j) < ↑j →
        GCG24UserItemFairness.OpposingTypes.pairShare (1 / 2) v j < 1 / 2) ∧
      (↑j = ↑(GCG24UserItemFairness.OpposingTypes.reverseItem j) →
        GCG24UserItemFairness.OpposingTypes.pairShare (1 / 2) v j = 1 / 2)
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_e_lemma16_midpoint_order_algebra

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 30

Source locator: cited publication, lines 3845-3925

Verbatim source input

Lemma 17. Let (x, z) be an optimal solution to Problem 11. For $j > n+1$
2 , $x_j = 0$.
In particular, if we let $h = \lfloor n+1$
2 $\rfloor$ indicate the index halfway through the set of items, this means that
we can simplify the problem above to
 $\arg \max$
 $x_1, \dots, x_h,$
 λ
 h
 $z_1, \dots, z_h,$
 λ
subject to $I_j(x, z) = \lambda, 1 \leq j \leq h$
 x
 x
 x
 $x_j =$
 $z_j +$
 $z_{n-j+1} = 1$
 $j \leq h$
 $j \leq h$
(12)
 $h < j \leq n$
 $x_j, z_j \geq 0$.
Now this is a linear program with $2h + 1$ variables and $h + 2$ constraints, so by an argument analogous
to Problem 12, we find that every basic feasible solution $x_1:h, z_1:h$ must share $h - 1$ zeros. This means
that there is a single index t such that $x_t > 0, z_t > 0$; if there were more than one, by the pigeonhole
principle there must be some $t' \leq h$ such that $x_{t'} = z_{t'} = x_{n-t'} + 1 = 0$, and $I_{t'}(x, z) = 0$. Then
*
 $I_{\min}(x, z) = 0$; however, by Lemma 1, $I_{\min}(x, z) = I_{\min}$
 > 0 , so we have a contradiction.
In terms of x, z ,
 $v + v_{n-j+1}$
 z_j
2
 $v_j + v_{n-j+1}$
 $\beta v_j + \beta v_{n-j+1} + (1 - 2\beta)$
2
 $v + v$
 $\beta v_j x_j + \beta v_{n-j+1} x_{n-j+1} + (1 - 2\beta) j_{n-j+1}$
 z_j
2
=
 $v_j + v_{n-j+1}$
2
 $I_j(x, z) =$
 $\beta v_j x_j + \beta v_{n-j+1} x_{n-j+1} + (1 - 2\beta) j$
[U+0012 control character]
= 2β
 v_j
 v_{n-j+1}
 $x_j +$
 x_{n-j+1}
 $v_j + v_{n-j+1}$
 $v_j + v_{n-j+1}$
41
[U+0013 control character]
+ $(1 - 2\beta)z_j$
[form-feed in source extraction]
 v
 j
. Note that q_j does not depend on β , and q_j is decreasing in j . Then
Let $q_j = v_j + v_{n-j+1}$
 $I_j(x, y, z) = 2\beta (q_j x_j + (1 - q_j) y_j) + (1 - 2\beta) z_j$.
Part 2. Moreover, the set of j such that $x_j = 0$ is contiguous; similarly for z_j .

Expanded PaperInterface specification

$\forall \{n : \mathbb{N}\} [\text{NeZero } n] \{\beta : \mathbb{R}\} \{v : \text{GCG24UserItemFairness.Item } n \rightarrow \mathbb{R}\} \{p : \text{GCG24UserItemFairness.TypePolicy } 3 \ n\}$
 $\{\ell : \mathbb{R}\},$

```

2 < n →
(↑n)-1 < beta →
beta < 1 / 2 →
(∀ (j : GCG24UserItemFairness.Item n), 0 < v j) →
GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v →
GCG24UserItemFairness.OpposingTypes.Theorem4MirrorSymmetricPolicy p →
(∀ (j : GCG24UserItemFairness.Item n),
GCG24UserItemFairness.OpposingTypes.theorem4Problem11PolicyItemValue beta v p j = ell) →
(∀ (p' : GCG24UserItemFairness.TypePolicy 3 n) (ell' : ℝ),
GCG24UserItemFairness.OpposingTypes.Theorem4MirrorSymmetricPolicy p' →
GCG24UserItemFairness.OpposingTypes.theorem4Problem11LPFeasible beta v p' ell' → ell' ≤ ell) →
∀ (j : GCG24UserItemFairness.Item n),
↑(GCG24UserItemFairness.OpposingTypes.reverseItem j) < ↑j → (p 0) j = 0

```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_e_lemma17_every_optimal_solution_has_no_right_half_support

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 31

Source locator: cited publication, lines 396-397

Verbatim source input

Theorem 3. πU
 $|I(\alpha)$ is decreasing in α for $0 < \alpha \leq 1/2$, and increasing in α for $1/2 \leq \alpha < 1$.

Expanded PaperInterface specification

```
∀ {m n : ℕ} [inst : NeZero m] [inst_1 : NeZero n] (S S' : GCG24UserItemFairness.RecommendationModel.SymmetricData m n 2)
(hTypes : Function.Surjective S.types.toType) (hTypes' : Function.Surjective S'.types.toType) {alpha alpha' : ℝ}
{v : GCG24UserItemFairness.Item n → ℝ},
(S.canonicalReductionOfSurjective hTypes).reduced = GCG24UserItemFairness.OpposingTypes.twoTypeReducedModel alpha v →
(S'.canonicalReductionOfSurjective hTypes').reduced =
  GCG24UserItemFairness.OpposingTypes.twoTypeReducedModel alpha' v →
  0 < alpha →
  alpha < 1 →
  0 < alpha' →
  alpha' < 1 →
  alpha ≤ alpha' →
  alpha ≤ 1 / 2 →
  alpha' ≤ 1 / 2 →
  (∀ (j : GCG24UserItemFairness.Item n), 0 < v j) →
  GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v →
  S'.model.priceOfFairness ≤ S.model.priceOfFairness
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.theorem3_price_decreases_first_half

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 32

Source locator: cited publication, lines 396-397

Verbatim source input

Theorem 3. π_U
 $|I(\alpha)$ is decreasing in α for $0 < \alpha \leq 1/2$, and increasing in α for $1/2 \leq \alpha < 1$.

Expanded PaperInterface specification

```
∀ {m n : ℕ} [inst : NeZero m] [inst_1 : NeZero n] (S S' : GCG24UserItemFairness.RecommendationModel.SymmetricData m n 2)
(hTypes : Function.Surjective S.types.toType) (hTypes' : Function.Surjective S'.types.toType) {alpha alpha' : ℝ}
{v : GCG24UserItemFairness.Item n → ℝ},
(S.canonicalReductionOfSurjective hTypes).reduced = GCG24UserItemFairness.OpposingTypes.twoTypeReducedModel alpha v →
(S'.canonicalReductionOfSurjective hTypes').reduced =
  GCG24UserItemFairness.OpposingTypes.twoTypeReducedModel alpha' v →
  0 < alpha →
  alpha < 1 →
  0 < alpha' →
  alpha' < 1 →
  alpha ≤ alpha' →
  1 / 2 ≤ alpha →
  1 / 2 ≤ alpha' →
  (∀ (j : GCG24UserItemFairness.Item n), 0 < v j) →
  GCG24UserItemFairness.OpposingTypes.StrictlyDecreasingByIndex v →
  S.model.priceOfFairness ≤ S'.model.priceOfFairness
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.theorem3_price_increases_second_half

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 33

Source locator: cited publication, lines 456-476

Verbatim source input

Theorem 4. If $\beta > n1$ and w and $\hat{w}$ are as described above, then fairness constraints can arbitrarily worsen the price of misestimation.

- The price of misestimation without fairness constraints is low: for all $\{v_j\}$, there is a

* recommendation policy $\hat{p}$ that solves the misestimated problem $U_{\min}(\theta, \hat{w})$ so that

$$\pi_U(\theta, w, \hat{w}) \leq$$

1
:
2

- The price of misestimation with fairness constraints can be arbitrarily large: $\forall \epsilon$, there exists

* $\{v_j\}$ and a recommendation policy $\hat{p}$ that solves the problem $U_{\min}(1, \hat{w})$ such that

$$\pi_U(1, w, \hat{w}) > 1 - \epsilon.$$

Expanded PaperInterface specification

```
∀ {m n : ℕ} [inst : NeZero m] [inst_1 : NeZero n] (E : GCG24UserItemFairness.EstimatedRecommendationModel m n)
  (Strue : GCG24UserItemFairness.RecommendationModel.SymmetricData m n 2)
  (Sest : GCG24UserItemFairness.RecommendationModel.SymmetricData m n 3)
  (hTypesTrue : Function.Surjective Strue.types.toType) (hTypesEst : Function.Surjective Sest.types.toType)
  {beta eps : ℝ} (u : GCG24UserItemFairness.User m),
  GCG24UserItemFairness.assumption_theorem4_true_model_reduction E Strue →
    GCG24UserItemFairness.assumption_theorem4_estimated_model_reduction E Sest →
    (Strue.canonicalReductionOfSurjective hTypesTrue).reduced =
      GCG24UserItemFairness.OpposingTypes.twoTypeReducedModel (1 / 2)
      (GCG24UserItemFairness.OpposingTypes.theorem4SmallValueVector eps) →
    (Sest.canonicalReductionOfSurjective hTypesEst).reduced =
      GCG24UserItemFairness.OpposingTypes.theorem4EstimatedReducedModel beta
      (GCG24UserItemFairness.OpposingTypes.theorem4SmallValueVector eps) →
    (∀ (u : GCG24UserItemFairness.User m), Sest.types.toType u = 0 → Strue.types.toType u = 0) →
    (∀ (u : GCG24UserItemFairness.User m), Sest.types.toType u = 1 → Strue.types.toType u = 1) →
      Strue.types.toType u = 1 →
        Sest.types.toType u = 2 →
          0 < eps →
            (↑n)-1 < beta →
              (have p0 :=
                GCG24UserItemFairness.OpposingTypes.theorem4NoFairnessPolicyCollapsed
                (GCG24UserItemFairness.OpposingTypes.theorem4SmallValueVector eps);
                E.SolvesEstimatedProblem 0 ((Sest.canonicalReductionOfSurjective hTypesEst).liftedPolicy p0) ∧
                E.priceOfMisestimation 0 ((Sest.canonicalReductionOfSurjective hTypesEst).liftedPolicy p0) ≤
                1 / 2) ∧
                ∃ p1,
                E.SolvesEstimatedProblem 1 ((Sest.canonicalReductionOfSurjective hTypesEst).liftedPolicy p1) ∧
                1 - eps <
                E.priceOfMisestimation 1 ((Sest.canonicalReductionOfSurjective hTypesEst).liftedPolicy p1)
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.theorem4_misestimation_tradeoff_typeOne

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 34

Source locator: cited publication, lines 456-476

Verbatim source input

Theorem 4. If $\beta > n_1$ and w and $\hat{w}$ are as described above, then fairness constraints can arbitrarily worsen the price of misestimation.

- The price of misestimation without fairness constraints is low: for all $\{v_j\}$, there is a

* recommendation policy $\hat{p}$ that solves the misestimated problem $U_{\min}(\theta, \hat{w})$ so that

$$\pi_U(\theta, w, \hat{w}) \leq$$

1
:
2

- The price of misestimation with fairness constraints can be arbitrarily large: $\forall \epsilon$, there exists

* $\{v_j\}$ and a recommendation policy $\hat{p}$ that solves the problem $U_{\min}(1, \hat{w})$ such that

$$\pi_U(1, w, \hat{w}) > 1 - \epsilon.$$

Expanded PaperInterface specification

```
∀ {m n : ℕ} [inst : NeZero m] [inst_1 : NeZero n] (E : GCG24UserItemFairness.EstimatedRecommendationModel m n)
  (Strue : GCG24UserItemFairness.RecommendationModel.SymmetricData m n 2)
  (Sest : GCG24UserItemFairness.RecommendationModel.SymmetricData m n 3)
  (hTypesTrue : Function.Surjective Strue.types.toType) (hTypesEst : Function.Surjective Sest.types.toType)
  {beta eps : ℝ} (u : GCG24UserItemFairness.User m),
  GCG24UserItemFairness.assumption_theorem4_true_model_reduction E Strue →
    GCG24UserItemFairness.assumption_theorem4_estimated_model_reduction E Sest →
    (Strue.canonicalReductionOfSurjective hTypesTrue).reduced =
      GCG24UserItemFairness.OpposingTypes.twoTypeReducedModel (1 / 2)
      (GCG24UserItemFairness.OpposingTypes.theorem4SmallValueVector eps) →
    (Sest.canonicalReductionOfSurjective hTypesEst).reduced =
      GCG24UserItemFairness.OpposingTypes.theorem4EstimatedReducedModel beta
      (GCG24UserItemFairness.OpposingTypes.theorem4SmallValueVector eps) →
    (∀ (u : GCG24UserItemFairness.User m), Sest.types.toType u = 0 → Strue.types.toType u = 0) →
    (∀ (u : GCG24UserItemFairness.User m), Sest.types.toType u = 1 → Strue.types.toType u = 1) →
    Strue.types.toType u = 0 →
    Sest.types.toType u = 2 →
    0 < eps →
    (↑n)-1 < beta →
    (have p0 :=
      GCG24UserItemFairness.OpposingTypes.theorem4NoFairnessPolicyCollapsed
      (GCG24UserItemFairness.OpposingTypes.theorem4SmallValueVector eps);
      E.SolvesEstimatedProblem 0 ((Sest.canonicalReductionOfSurjective hTypesEst).liftedPolicy p0) ∧
      E.priceOfMisestimation 0 ((Sest.canonicalReductionOfSurjective hTypesEst).liftedPolicy p0) ≤
      1 / 2) ∧
    ∃ p1,
      E.SolvesEstimatedProblem 1 ((Sest.canonicalReductionOfSurjective hTypesEst).liftedPolicy p1) ∧
      1 - eps <
      E.priceOfMisestimation 1 ((Sest.canonicalReductionOfSurjective hTypesEst).liftedPolicy p1)
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.theorem4_misestimation_tradeoff_typeZero

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 35

Source locator: cited publication, lines 456-476

Verbatim source input

Theorem 4. If $\beta > n_1$ and w and $\hat{w}$ are as described above, then fairness constraints can arbitrarily worsen the price of misestimation.

- The price of misestimation without fairness constraints is low: for all $\{v_j\}$, there is a

* recommendation policy $\hat{p}$ that solves the misestimated problem $U_{\min}(\theta, \hat{w})$ so that

$$\frac{M}{\pi U}(\theta, w, \hat{w}) \leq$$

$$\frac{1}{2}$$

- The price of misestimation with fairness constraints can be arbitrarily large: $\forall \epsilon$, there exists

* $\{v_j\}$ and a recommendation policy $\hat{p}$ that solves the problem $U_{\min}(1, \hat{w})$ such that

$$\frac{M}{\pi U}(1, w, \hat{w}) > 1 - \epsilon.$$

Expanded PaperInterface specification

```
∀ {m n : ℕ} [inst : NeZero m] [inst_1 : NeZero n] (E : GCG24UserItemFairness.EstimatedRecommendationModel m n)
  (Strue : GCG24UserItemFairness.RecommendationModel.SymmetricData m n 2)
  (Sest : GCG24UserItemFairness.RecommendationModel.SymmetricData m n 3)
  (hTypesTrue : Function.Surjective Strue.types.toType) (hTypesEst : Function.Surjective Sest.types.toType) {beta : ℝ}
  {v : GCG24UserItemFairness.Item n → ℝ},
  GCG24UserItemFairness.assumption_theorem4_true_model_reduction E Strue →
    GCG24UserItemFairness.assumption_theorem4_estimated_model_reduction E Sest →
      (Strue.canonicalReductionOfSurjective hTypesTrue).reduced =
        GCG24UserItemFairness.OpposingTypes.twoTypeReducedModel (1 / 2) v →
        (Sest.canonicalReductionOfSurjective hTypesEst).reduced =
          GCG24UserItemFairness.OpposingTypes.theorem4EstimatedReducedModel beta v →
          (∀ (u : GCG24UserItemFairness.User m), Sest.types.toType u = 0 → Strue.types.toType u = 0) →
          (∀ (u : GCG24UserItemFairness.User m), Sest.types.toType u = 1 → Strue.types.toType u = 1) →
            (↑n)-1 < beta →
              GCG24UserItemFairness.assumption_theorem4_universal_value_vector v →
                have p := GCG24UserItemFairness.OpposingTypes.theorem4NoFairnessPolicyCollapsed v;
                E.SolvesEstimatedProblem 0 ((Sest.canonicalReductionOfSurjective hTypesEst).liftedPolicy p) ∧
                E.priceOfMisestimation 0 ((Sest.canonicalReductionOfSurjective hTypesEst).liftedPolicy p) ≤ 1 / 2
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.theorem4_misestimation_without_fairness_universal

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 36

Source locator: cited publication, lines 149-150

Verbatim source input

Given a recommendation policy p , user fairness is quantified as the minimum normalized user utility $U_{\min}(p, w) = \min_i U_i(p, w)$, and item fairness analogously as $I_{\min}(p, w) = \min_j I_j(p, w)$.

Expanded PaperInterface specification

```
∀ {m n : ℕ} [inst : NeZero m] [inst_1 : NeZero n] (W : GCG24UserItemFairness.RecommendationModel m n)
  (p : GCG24UserItemFairness.Policy m n),
  GCG24UserItemFairness.ProofBridge.userFairness W p =
    EconCSLib.finiteMin (GCG24UserItemFairness.ProofBridge.normalizedUserUtility W p)
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.userFairness_realizes_spec

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 37

Source locator: cited publication, lines 1759-1800

Verbatim source input

Lemma 2. Let F denote the feasible set of Problem 1,
 $F \triangleq \arg \max$
 $\min (\rho) \triangleq \arg \max$
 $\min I_j (\rho).$
 m
 m
 $\rho \in \Delta_{n-1}$
 $\rho \in \Delta_{n-1}$
Then F is the solution set of the linear program
 $F = \arg \max$
 m
 $\rho \in \Delta_{n-1}, \lambda$
 j
(3)
 λ
(4)
subject to $I_j (\rho) = \lambda$
 $\forall j.$
Now, condition (iii) on S implies that there is a unique ρ^* in $S \cap F$. Thus
 $\rho^* = F \cap S = \arg$
 $\max$
 λ
 m
 $\rho \in \Delta_{n-1} \cap S, \lambda$
.
(5)
subject to $I_j (\rho) = \lambda \forall j,$
We can add S as a constraint in Problem 4 because we know that there is at least one solution to
Problem 4 in S – namely, ρ^* – so this additional constraint will not change the optimal objective
value, but will constrain the set of optimal solutions to be in S , as desired.
By condition (ii), ρ^* is an optimal solution for Problem 1; by condition (i), S can be described by a
finite set of linear constraints, and thus Problem 5 is a linear program; call this L .

Expanded PaperInterface specification

```
∀ {m n : ℕ} [NeZero m] [inst : NeZero n] (W : GCG24UserItemFairness.RecommendationModel m n),
  W.Positive →
    ∀ (rho : GCG24UserItemFairness.Policy m n),
      W.itemFairness rho = W.optimalItemFairness ↔
        ∃ ell,
          W.itemFairnessEqualityLPFeasible rho ell ∧
            ∀ (rho' : GCG24UserItemFairness.Policy m n) (ell' : ℝ),
              W.itemFairnessEqualityLPFeasible rho' ell' → ell' ≤ ell
```

Lean proof endpoint (built separately)

GCG24UserItemFairness.PaperInterface.appendix_c_lemma2_item_fairness_equality_lp_solution_set

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation